

WIT Standard Communication Protocol

Instructions for Command:

Command

The serial port sending command must be completed within 10S, otherwise it will be automatically locked. To avoid this situation, please do the following steps first.

1. Enter the unlock command (0XFF 0XAA 0X69 0X88 0XB5, 0X means hexadecimal)
2. Enter the command that needs to modify or read the data
3. Save command (0XFF 0XAA 0X00 0X00 0X00)

Register

ADDRESS (Hex)	ADDRESS (Hex)	REGISTER NAME	FUNCTION	SERIAL I/F	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit9	Bit8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
00	00	SAVE	Save/Reset	R/W	SAVE[15:0]															
01	01	CALSW	Calibration Mode	R/W													CALSW[3:0]			
02	02	RSW	Output Content	R/W						GS A	QU A T E R	VE L O C I T Y	G P S	P R E S S	P O R T	M A G	A N G L E	G Y R O	A C C	T I M E
03	03	RRATE	Output Rate														RRATE[3:0]			

1 A	2 6	IICADD R	Device address	R/ W															IICADDR[7:0]
	3 7	LEDOF F	Turn off LED light	R/ W															LE D O F F
	4 8	MAGR AN GX	Magnetic field X calibrati on range	R/ W	MAGRANGX[15:0]														
	5 9	MAGR AN GY	Magnetic field Y calibrati on range	R/ W	MAGRANGY[15:0]														
	6 A	MAGR AN GZ	Magnetic field Z calibrati on range	R/ W	MAGRANGZ[15:0]														
1 F	3 1	BAND WID TH	Bandwid th	R/ W															BANDWIDTH[3: 0]
2 0	3 2	GYRO RA NGE	Gyro ra nge																GYRORANGE[3: 0]
2 1	3 3	ACCR AN GE	Accelerati on range	R/ W															ACCRANGE[3:0]
2 2	3 4	SLEEP	Sleep	R/ W															SL EE P
2 3	3 5	ORIEN T	Installati on direction	R/ W															O RI E N T

3 8	5 6	GY	Angular velocity Y	R	GY[15:0]
3 9	5 7	GZ	Angular velocity Z	R	GZ[15:0]

3 A	5 8	HX	Magnetic field X	R	HX[15:0]
3 B	5 9	HY	Magnetic field Y	R	HY[15:0]
3 C	6 0	HZ	Magnetic field Z	R	HZ[15:0]
3 D	6 1	Roll	Roll	R	Roll[15:0]
3 E	6 2	Pitch	Pitch angle	R	Pitch[15:0]
3 F	6 3	Yaw	Heading	R	Yaw[15:0]
4 0	6 4	TEMP	Tempera tur e	R	TEMP[15:0]
4 1	6 5	D0Statu s	D0 pin state	R	D0Status[15:0]
4 2	6 6	D1Statu s	D1pin state	R	D1Status[15:0]
4 3	6 7	D2Statu s	D2 pin state	R	D2Status[15:0]
4 4	6 8	D3Statu s	D3 pin state	R	D3Status[15:0]
4 5		PressureL	Air pressure low 16 bits	R	PressureL[15:0]
4 6		PressureH	Air pressure high16 bits	R	PressureH[15:0]
4 7	7 1	HeightL	Height lower 16 bits	R	HeightL[15:0]

4 8	7 2	HeightH	Height high 16 bits	R	HeightH[15:0]
4 9	7 3	LonL	Longitud e lower 16 bits	R	LonL[15:0]
4 A	7 4	LonH	Longitud e	R	LonH[15:0]

			high 16 bits		
4 B	7 5	LatL	Latitude lower 16 bits	R	LatL[15:0]
4 C	7 6	LatH	Latitude high 16 bits	R	LatH[15:0]
4 D	7 7	GPSHe igh t	GPS Altitude	R	GPSHeight[15:0]
4 E	7 8	GPSYA W	GPS Heading	R	GPSYAW[15:0]
4 F	7 9	GPSVL	GPS round speed low 16 bits	R	GPSVL[15:0]
5 0	8 0	GPSVH	GPS ground speed high 16 bits	R	GPSVH[15:0]
5 1	8 1	q0	Quaterni on 0	R	q0[15:0]
5 2	8 2	q1	Quaternio n1	R	q1[15:0]
5 3	8 3	q2	Quaternio n2	R	q2[15:0]
5 4	8 4	q3	Quaternio n3	R	q3[15:0]

55	85	SVNUM	Number of satellites	R	SVNUM[15:0]			
56	86	PDOP	Position accuracy	R	PDOP[15:0]			
57	87	HDOP	Horizontal accuracy	R	HDOP[15:0]			
58	88	VDOP	Vertical accuracy	R	VDOP[15:0]			

59	89	DELAYT	Alarm delay	R	DELAYT[15:0]			
60	90	XMIN	X-axis angle alarm min.	R/W	XMIN[15:0]			
61	91	XMAX	X-axis angle alarm max.	R/W	XMAX[15:0]			
62	92	BATVAL	Supply voltage	R	BATVAL[15:0]			
63	93	ALARM PIN MAPPING	Alarm Pin Mapping	R/W	X-ALARM[15:12]	X+ALARM[11:8]	Y-ALARM[7:4]	Y+ALARM[3:0]
64	94	YMIN	Y-axis alarm min.	R/W	YMIN[15:0]			
65	95	YMAX	Y-axis alarm max.	R/W	YMAX[15:0]			
66	96	GYROCALITHR	Gyro Still Threshold	R/W	GYROCALITHR[15:0]			

1	2	YREF PITCH	Pitch angle zero reference value	R	YREFPITCH[15:0]	
5	7	NUMBER D1	Device No1- 2	R	ID2[15:8]	ID1[7:0]
9	8	NUMBER D2	Device No3- 4	R	ID4[15:8]	ID3[7:0]
13	9	NUMBER D3	Device No5- 6	R	ID6[15:8]	ID5[7:0]
17	10	NUMBER D4	Device No7- 8	R	ID8[15:8]	ID7[7:0]
21	11	NUMBER D5	Device No9- 10	R	ID10[15:8]	ID9[7:0]
25	12	NUMBER D6	Device No11-12	R	ID12[15:8]	ID11[7:0]

Protocol Format

Read

- The Data is sent in hexadecimal, not ASCII.
- Each data is transmitted in sequence by low byte and high byte, and the two are combined into a signed short type of data. For example, for data DATA1, DATA1L is the low byte and DATA1H is the high byte. The conversion method is as follows:

Suppose DATA1 is the actual data, DATA1H is its high-byte part, DATA1L is its low-byte part.

Then: $DATA1 = (\text{short})((\text{short})DATA1H \ll 8 | DATA1L)$. It must be noted here that DATA1H needs to be coerced into a signed short type of data before shifting, and the data type of DATA1 is also a signed short type, so that it can represent negative numbers.

Pro toc ol he ad er	Da ta co nt e nt	Data lower 8 bits	Data higher 8 bits	Data lower 8 bits	Data higher 8 bits	Data lower 8 bits	Data higher 8 bits	Data lower 8 bits	Data higher 8 bits	SU MC RC
0x55	TY P E 【 1 】	DATA 1L[7:0]	DATA1 H[1 5:8]	DATA 2L[7:0]	DATA2 H[1 5:8]	DATA 3L[7:0]	DATA3 H[1 5:8]	DATA 4L[7:0]	DATA4 H[1 5:8]	SU MC RC 【2 】

【1】TYPE(Data content):

TYPE	Remark
0x50	Time
0x51	Acceleration
0x52	Angular velocity
0x53	Angle

0x54	Magnetic field
0x55	Port
0x56	Barometric altitude
0x57	Latitude and Longitude
0x58	ground speed
0x59	Quaternion
0x5A	GPS Location accuracy

0x5F	Read
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【2】SUMCRC(Data and calibration):

SUMCRC=0x55+TYPE+DATA1L+DATA1H+DATA2L+DATA2H+DATA3L+DATA3H+DATA4L+DATA4H

SUMCRC is a char type, taking the lower 8 bits of the checksum

Time Output

0x55	0x50	YY	MM	DD	HH	MN	SS	MSL	MSH	SUM
Name	Describe		Remark							
YY	Year									
MM	Month									
DD	Day									
HH	Hour									
MN	Minute									
SS	Seconds									
MSL	MS lower 8 bits		Millisecond calculation formula: milliseconds=((MSH<<8) MSL)							
MSH	MS higher 8 bits									
SUM	Checksum		SUM=0x55+0x50+YY+MM+DD+HH+MN+SS+MSL+MSH							

Acceleration Output

0x55	0x51	AxL	AxH	AyL	AyH	AzL	AzH	TL	TH	SUM
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Nam e	Description	Remark
AxL	Accelerati on X low 8 bits	Acceleration $X=((AxH<8) AxL)/32768*16g$ (g is the acceleration of gravity, preferably 9.8m/s ²)
AxH	Accelerati on X high 8 bits	
AyL	Accelerati on Y low 8 bits	Acceleration $Y=((AyH<8) AyL)/32768*16g$ (g is the acceleration of gravity, preferably 9.8m/s ²)
AyH	Accelerati on Yhigh 8 bits	
AzL	Accelerati on Z low 8 bits	Acceleration $Z=((AzH<8) AzL)/32768*16g$ (g is the acceleration of gravity, preferably 9.8m/s ²)
AzH	Accelerati on Z high 8 bits	
TL	8-bit lower temperat ure	Temperature calculation formula: Temperature= $((TH<8) TL) /100\text{ }^{\circ}\text{C}$
TH	8-bit higher temperat ure	
SUM	Check sum	SUM=0x55+0x51+AxL+AxH+AyL+AyH+AzL+AzH+TL+Th

Angular velocity output

0x55	0x52	WxL	WxH	WyL	WyH	WzL	WzH	VolL	VolH	SUM
Name	Description									
WxL	Angular velocity X low 8 bits	Angular velocity X=((WxH<<8) WxL)/32768*2000°/s								
WxH	Angular velocity X high 8 bits									
WyL	Angular velocity Y low 8 bits	Angular velocity Y=((WyH<<8) WyL)/32768*2000°/s								
WyH	Angular velocity Y high 8 bits									
WzL	Angular velocity Z low 8 bits	Angular velocity Z=((WzH<<8) WzL)/32768*2000°/s								
WzH	Angular velocity Z high 8 bits									
VolL	Voltage lower 8 bits	(Non-Bluetooth products, the data is invalid) Voltage								

	bits	
D1L	D1 status lower 8 bits	D1status=((D1H<<8) D1L)
D1H	D1 status higher 8 bits	
D2L	D2 status lower 8 bits	D2 status=((D2H<<8) D2L)
D2H	D2 status higher 8 bits	
D3L	D3 status lower 8 bits	D3 status=((D3H<<8) D3L)
D3H	D3 status higher 8 bits	
SUM	Check Sum	SUM=0x55+0x54+D0L+D0H+D1L+D1H+D2L+D2H+D3L+D3H

explanation:

- When the port mode is set to analog input, the port status data represents the analog voltage. The actual voltage is calculated according to the formula below:

$$U = D \times \text{Status} / 1024 \times U_{\text{vcc}}$$

- U_{vcc} is the power supply voltage of the chip. Since there are LDOs on the chip, if the power supply voltage of the module is greater than 3.5V, U_{vcc} is 3.3V. If the

module power supply voltage is less than 3.5V, $U_{vcc} = \text{power supply voltage} - 0.2V$.

- When the port mode is set to digital input, the port status data indicates the digital level status of the port, high level is 1 and low level is 0.
- When the port mode is set to high-level output mode, the port status data is 1.
- When the port mode is set to low-level output mode, the port status data bit is 0.

Air pressure altitude output

0x55	0x56	P0	P1	P2	P3	H0	H1	H2	H3	SUM
Name	Description	Remark								
P0	Air pressure[7:0]	Air pressure=[(P3<<24) (P2<<16) (P1<<8) P0](Pa)								
P1	Air pressure[[15:8]									
P2	Air pressure[[23:16]									
P3	Air pressure[31:24]									
H0	Altitude[7:0]	Altitude=(H3<<24) (H2<<16) (H1<<8) H0(cm)								
H1	Altitude[15:8]									
H2	Altitude[23:16]									
H3	Altitude[31:24]									
SUM	Check sum	SUM=0x55+0x56+P0+P1+P2+P3+H0+H1+H2+H3								

Latitude and longitude

0x55	0x57	Lon0	Lon1	Lon2	Lon3	Lat0	Lat1	Lat2	Lat3	SUM
Name	Description	Remark								
	itude[7:0]	Longitude=(Lon3<<24) (Lon2<<16) (Lon1<<8) Lon0								
	itude[15:8]									
Lon2	Longitude[23:16]									
Lon3	Longitude[31:24]									
Lat0	Latitude[7:0]	Latitude=(Lat3<<24) (Lat2<<16) (Lat1<<8) Lat0								
Lat1	Latitude[15:8]									
Lat2	Latitude[23:16]									
Lat3	Latitude[31:24]									
	SUM Checksum	SUM=0x55+0x57+Lon0+Lon1+Lon2+Lon3+Lat0+Lat1+Lat2+ Lat3								

Explanation:

The NMEA8013 standard stipulates that the longitude output format of GPS is dd mm.mmmmm (dd is degrees, mm.mmmmm is minutes), and the decimal point is removed from the output of longitude/latitude, so the degrees of longitude/latitude can be calculated as follows:

dd=Lon [31:0]/10000000;

dd=Lat [31:0]/10000000;

Longitude/latitude fraction calculation:

mm.mmmmm= (Lon [31:0] %10000000)/100000(% indicates the remainder operation)

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mm.mmmmm=(Lat [31:0] %10000000)/100000(% indicates the remainder operation)
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GPS data output

0x555	0x58	GPS Height L	GPS Height H	GPS Yaw L	GPS Yaw H		GP GP				
Name	Description	Remark									
GPS Height L	GPS altitude[7:0]	GPS altitude=((GPSHeightH<<8) GPSHeightL)/10(m)									
GPS Height H	GPS altitude[15:8]										
GPS Yaw L	GPS heading [7:0]	GPS heading =((GPSYawH<<8) GPSYawL)/100(°)									

GPS Yaw with	GPS Heading [15:8]	
GPS Velocity 0	GPS Ground Speed d[7:0]	GPS Ground Speed = ((GPSV3<<24) (GPSV2<<16) (GPSV1<<8) GPSV0)/1000(km/h)
GPS Velocity 1	GPS Ground d	

	Speed d[15:8]	
GPS Velocity 2	GPS Ground Speed d[23:16]	
GPS Velocity 3	GPS Ground Speed d[31:24]	

SNL	Satellite number low 8 bits	GPSSatellite number=((SNH<<8) SNL)
SNH	Satellite number high 8 bits	
PDOPL	Position accuracy low 8 bits	Position accuracy=((PDOPH<<8) PDOPL)/100
PDOPH	Position accuracy high 8 bits	
HDOPL	Horizontal precision low 8 bits	Horizontal precision=((HDOPH<<8) HDOPL)/100

R E G 1L	Regist er 1 low 8 bits	REG1[15:0]=((REG1H<<8) REG1L)
R E G 1 H	Regist er 1 high 8 bits	
R E G 2L	Regist er 2 low 8 bits	REG2[15:0]=((REG2H<<8) REG2L)
R E G 2 H	Regist er 2 high 8 bits	
R E G 3L	Regis ter 3 low 8 bits	REG3[15:0]=((REG3H<<8) REG3L)
R E G 3 H	Regist er 3 high 8 bits	

R E G 4L	Regist er 4 low 8 bits	REG4[15:0]=((REG4H<<8) REG4L)
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R E G 5 H	Regist er 4 high 8 bits	
SU M	Chec ks um	SUM=0x55+0x5F+REG1L+REG1H+REG2L+REG2H+REG3L+REG 3H+REG4L+REG4H

Example:

Read register "AXOFFSET", return: 0x55 0x5F AXOFFSET [7:0] AXOFFSET [15:8]
 AYOFFSET [7:0] AYOFFSET [15:8] AZOFFSET [7:0] AZOFFSET [15:8] GXOFFSET [7:0]
 GXOFFSET [15:8] SUM

Write Format

- The following data are all Hex codes in hexadecimal
- All settings need to operate the unlock register first(KEY)

Head	Head	Register	Data lower 8 bits	Data high 8 bits
0xFF	0xAA	ADDR	DATAL[7:0]	DATAH[15:8]

- The data is sent in hexadecimal not ASCII.
- Each data is divided into low byte and high byte and transmitted in turn, and the two are combined into a signed short data.

For example, DATA, DATAL is the low byte, DATAH is the high byte.

Conversion method: Assuming that DATA is the actual data, DATAH is its high-byte part, and DATAL is its low-byte part, then:DATA=(short)((short)DATAH<<8|DATAL)

DATAH must be cast to a signed short type before shifting, and the data type of DATA is also a signed short type, so that negative numbers can be represented.

Note:

There are three steps to perform the command write operation.

Step1. Unlock 0xFF 0xAA 0x69 0x88 0xB5

Step2. Send the command to be modified

Step3. Save 0xFF 0xAA 0x00 0x00 0x00

The flow chart is below. The data is sent in hexadecimal not ASCII.

Each data is divided into low byte and high byte and transmitted in turn, and the two are combined into a signed short data.

For example, DATA, DATAL is the low byte, DATAH is the high byte.

Conversion method: Assuming that DATA is the actual data, DATAH is its high-byte part, and DATAL is its low-byte part, then: $DATA = (\text{short})(\text{short})DATAH \ll 8 | DATAL$

DATAH must be cast to a signed short type before shifting, and the data type of DATA is also a signed short type, so that negative numbers can be represented.

Note:

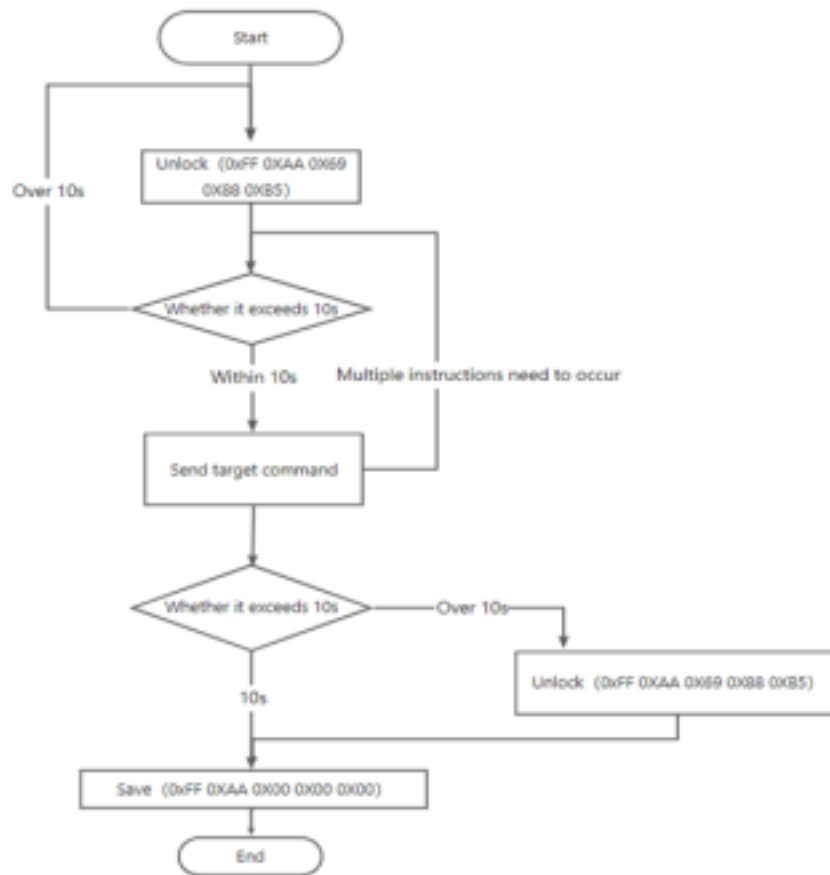
There are three steps to perform the command write operation.

Step1. Unlock 0xFF 0xAA 0x69 0x88 0xB5

Step2. Send the command to be modified

Step3. Save 0xFF 0xAA 0x00 0x00 0x00

The flow chart is below.



SAVE(Save/restart/reset)

Register Name: SAVE

Register Address: 0 (0x00)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Restart: 0x00FF Factory reset: 0x0001

Example: FF AA 00 FF 00 (Restart)

CALSW (Calibration mode)

Register Name: CALSW Register Address: 1 (0x01) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	CAL[3:0]	Set calibration mode: 0000(0x00): Normal working mode 0001(0x01): Automatic accelerometer calibration 0011(0x03): Height reset 0100(0x04): Set the heading angle to zero 0111(0x07): Magnetic Field Calibration (Spherical Fitting) 1000 (0x08): Set the angle reference 1001(0x09): Magnetic Field Calibration (Dual Plane Mode)
Example: FF AA 01 04 00 (Set heading angle to zero)		

RSW (Output content)

Register Name: RSW Register Address: 2 (0x02) Read and write direction: R/W Default: 0x001E		
Bit	NAME	FUNCTION

15:11		
10	GSA (0x5A)	0: off 1: on
9	QUATER (0x59)	0: off 1: on
8	VELOCITY (0x58)	0: off 1: on
7	GPS (0x57)	0: off 1: on
6	PRESS (0x56)	0: off 1: on
5	PORT (0x55)	0: off 1: on
4	MAG (0x54)	0: off 1: on
3	ANGLE (0x53)	0: off 1: on
2	GYRO (0x52)	0: off 1: on
1	ACC (0x51)	0: off 1: on
0	TIME (0x50)	0: off 1: on
Example: FF AA 02 3E 00 (set to only output acceleration, angular velocity, angle, magnetic field, port state)		

RRATE (Output rate)

Register Name: RRATE		
Register Address: 3 (0x03)		
Read and write direction: R/W		
Default: 0x0006		
Bit	NAME	FUNCTION
15:4		

3:0	RRATE[3:0]	Set the output rate: 0001(0x01): 0.2Hz 0010(0x02): 0.5Hz 0011(0x03): 1Hz 0100(0x04): 2Hz 0101(0x05): 5Hz 0110(0x06): 10Hz 0111(0x07): 20Hz 1000(0x08): 50Hz 1001(0x09): 100Hz 1011(0x0B): 200Hz 1011(0x0C): single return 1100(0x0D): no return
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Example: FF AA 03 03 00 (set 1Hz output)

Note: HWT906, WT931 can output 500Hz, 1000Hz

FF AA 03 0C 00: 500HzFF AA 03 0D 00: 1000Hz

FF AA 03 10 00 : single return

BAUD (Serial baud rate)

Register Name: BAUD Register Address: 4 (0x04) Read and write direction: R/W Default: 0x0002		
Bit	NAME	FUNCTION
15: 4		
3:0	BAUD[3:0]	Set the serial port baud rate: 0001(0x01): 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps 0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps 1000(0x08): 460800bps (Only supportsWT931/JY931/HWT606/HWT906) 1001(0x09): 921600bps (Only supportsWT931/JY931/HWT606/HWT906)
Example: FF AA 04 06 00 (Set serial port baud rate115200)		

AXOFFSET~HZOFFSET (Zero bias setting)

Register Name: AXOFFSET~HZOFFSET

Register Address: 5~13 (0x05~0x0D)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	AXOFFSET[15:0]	Acceleration X-axis zero offset, actualvalue=AXOFFSET[15:0]/10000(g)
15:0	AYOFFSET[15:0]	Acceleration Y-axis zero offset, actualvalue=AYOFFSET[15:0]/10000(g)
15:0	AZOFFSET[15:0]	Acceleration Z-axis zero offset, actualvalue=AZOFFSET[15:0]/10000(g)
15:0	GXOFFSET[15:0]	Angular velocity X-axis zero offset, actual value=GXOFFSET[15:0]/10000(°/s)
15:0	GYOFFSET[15:0]	Angular velocity Y-axis zero offset, actual value=GYOFFSET[15:0]/10000(°/s)
15:0	GZOFFSET[15:0]	Angular velocity Z-axis zero offset, actual value=GZOFFSET[15:0]/10000(°/s)
15:0	HXOFFSET[15:0]	Magnetic field X-axiszero offset
15:0	HYOFFSET[15:0]	Magnetic field Y-axiszero offset
15:0	HZOFFSET[15:0]	Magnetic field Z-axiszero offset

Example: FF AA 05 E8 03 (set acceleration X-axis offset 0.1g),0x03E8=1000
 , 1000/10000=0.1(g)

D0MODE~D3MODE (Port Mode Settings)

Register Name: D0MODE~D3MODE

Register Address: 14~17 (0x0E~0x11)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
3:0	D0MODE[3:0]	Set D0 port mode 0000(0x00): Analog input (default) 0001 (0x01): Digital input 0010 (0x02): Output digital high level 0011(0x03): Output digital low level
3:0	D1MODE[3:0]	Set D1 port mode 0000(0x00): Analog input (default) 0001(0x01): Digital input 0010(0x02): Output digital high level 0011(0x03): Output digital low level 0101(0x05): Set relative pose
3:0	D2MODE[3:0]	Set D2 port mode

		0000(0x00): Analog input (default) 0001(0x01): Digital input 0010(0x02): Output digital high level 0011(0x03): Output digital low level
3:0	D3MODE[3:0]	Set D3 port mode 0000(0x00): Analog input (default) 0001(0x01): Digital input 0010(0x02): Output digital high level 0011(0x03): Output digital low level
Example: FF AA 0E 03 00 (set D0 as the output digital low level mode)		

IICADDR (Device address)

Register Name: IICADDR Register Address: 26 (0x1A) Read and write direction: R/W Default: 0x0050		
Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address for I2C and Modbus communication 0x01~0x7F
Example: FF AA 1A 02 00 (set the device address to 0x02)		

LEDOFF

Register Name: LEDOFF		
Register address: 27 (0x1B)		
Read and write direction: R/W		
Default value: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	LEDOFF	1: Turn off the LED light 0: Turn on the LED light
Example: FF AA 1B 01 00 (turn off the LED light)		

MAGRANGX~MAGRANGZ

Register Name: MAGRANGX~MAGRANGZ		
Register Address: 28~30 (0x1C~0x1E)		
Read and write direction: R/W		
Default: 0x01F4		
Bit	NAME	FUNCTION
15:0	MAGRANGX[15:0]	Magnetic field calibration X-axis range
15:0	MAGRANGY[15:0]	Magnetic Field Calibration Y-axis range
15:0	MAGRANGZ[15:0]	Magnetic field calibration Z-axis range
Example: FF AA 1C F4 01 (set the calibration X-axis range to 500)		

BANDWIDTH

Register Name: BANDWIDTH		
Register Address: 31 (0x1F)		
Read and write direction: R/W		
Default: 0x0004		
Bit	NAME	FUNCTION
15:4		
3:0	BANDWIDTH[3:0]	Set bandwidth 0000(0x00): 256Hz 0001(0x01): 188Hz 0010(0x02): 98Hz 0011(0x03): 42Hz 0100(0x04): 20Hz 0101(0x05): 10Hz 0110(0x06): 5Hz
Example: FF AA 1F 01 00 (set the bandwidth to 188Hz)		

GYRORANGE

Register Name: GYRORANGE		
Register Address: 32 (0x20)		
Read and write direction: R/W		
Default: 0x0003		
Bit	NAME	FUNCTION

15:4		
3:0	GYRORANGE[3:0]	Set the gyro range 0011(0x03): 2000°/s The default is 2000°/s, fixed and cannot be set
Example: FF AA 20 03 00 (set the gyro range to 2000°/s)		

ACCRANGE

Register Name: ACCRANGE Register Address: 33 (0x21) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	ACCRANGE[3:0]	Set the accelerometer range 0000(0x00): ±2g 0011(0x03): ±16g This parameter cannot be set. The product's internal adaptive acceleration range will automatically switch to 16g when the acceleration exceeds 2g.
Example: FF AA 21 00 00 (set the accelerometer range to 16g)		

SLEEP

Register Name: SLEEP Register Address: 34 (0x22) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	SLEEP	set hibernate 1(0x01): sleep Any serial data to wake up
Example: FF AA 22 01 00 (go to sleep)		

ORIENT (Installation direction)

Register Name: ORIENT Register Address: 35 (0x23) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	ORIENT	Set the installation direction 0 (0x00): horizontal installation 1(0x01): Vertical installation (the Y-axis arrow of the coordinate axis must be upwards)

Example: FF AA 23 01 00 (set vertical installation)

AXIS6 (Algorithm)

Register Name: AXIS6

Register Address: 36 (0x24)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:1		
0	AXIS6	set algorithm 0(0x00): 9-axis algorithm (magnetic field solution heading angle, absolute heading angle) 1 (0x01): 6-axis algorithm (integral solution heading angle, relative heading angle)

Example: FF AA 24 01 00 (set 6-axis algorithm mode)

FILTK (K-value filtering)

Register Name: FILTK

Register Address: 37 (0x25)

Read and write direction: R/W

Default: 0x001E

Bit	NAME	FUNCTION
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15:0	FILTK[15:0]	Range: 1~10000, default 30 (modification is not recommended) The smaller the FILTK [15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the FILTK[15:0], the weaker the seismic performance and the stronger the real-time performance.
Example: FF AA 25 1E 00 (set K value filter to 30)		

GPSBAUD (GPS Baud Rate)

Register Name: GPSBAUD Register Address: 38 (0x26) Read and write direction: R/W Default: 0x0002		
Bit	NAME	FUNCTION
15:4		
3:0	GPSBAUD[3:0]	Set GPS baud rate: 0001(0x01): 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps 0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps

Example: FF AA 26 02 00 (set GPS baud rate 9600)

READADDR (Read registers)

Register Name: READADDR

Register Address: 39 (0x27)

Read and write direction: R/W

Default: 0x00FF

Bit	NAME	FUNCTION
15:8		
7:0	READADDR[7:0]	Read register range: Please refer to "Register Table"

Example:

Send: FF AA 27 34 00 (read acceleration X axis 0x34)

Return: 55 5F AXL AXH AYL AYH AZL AZH GXL GXH SUM

For details, please refer to "Read Register Return Value" in the "Read Format" chapter

ACCFILT (Acceleration filtering)

Register Name: ACCFILT

Register Address: 42 (0x2A)

Read and write direction: R/W

Default: 0x01F4

Bit	NAME	FUNCTION
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15:0	ACCFILT[15:0]	Range: 1~10000, default 500 (It is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 500) The smaller the ACCFILT [15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the ACCFILT [15:0], the weaker the seismic performance and the stronger the real-time performance. This parameter is an empirical value and needs to be debugged according to different environments. For example, in a tractor environment: ACCFILT[15:0] can be adjusted to 100, the vibration of the tractor is serious, and the seismic performance needs to be improved.
Example: FF AA 2A F4 01 (set acceleration filter 500)		

POWONSEND (Power-on output)

Register Name: POWONSEND Register Address: 45 (0x2D) Read and write direction: R/W Default: 0x0001		
Bit	NAME	FUNCTION
15:4		

3:0	POWONSEND[3:0]	Set the command to start: 0000(0x00): Turn off power-on data output 0001(0x01): Turn on power-on data output
Example: FF AA 2D 00 00 (turn on power-on data output)		

VERSION (Version number)

Register Name: VERSION Register Address: 46 (0x2E) Read and write direction: R Default: none		
Bit	NAME	FUNCTION
15:0	VERSION[15:0]	Different products, different version numbers
Example: Send: FF AA 27 2E 00 (read version number, 0x27 means read, 0x2E is version number register) Return: 55 5F VL VH XX XX XX XX XX XX SUM VERSION[15:0]=(short)((((short)VH<<8) VL)		

YYMM~MS (Chip time)

Register Name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	Month
7:0	YYMM[7:0]	Year
15:8	DDHH[15:8]	Hour
7:0	DDHH[7:0]	Day
15:8	MMSS[15:8]	Seconds
7:0	MMSS[7:0]	Minute
15:0	MS[15:0]	Millisecond

Example:

FF AA 30 16 03 (set year 22-03)

FF AA 31 0C 09 (set hour 12-09)

FF AA 32 1E 3A (set minute seconds 30:58)

FF AA 33 F4 01 (set ms 500)

Example:

Send: FF AA 27 30 00 (read version number, 0x27 means read, 0x30 is year month register)

Return: 55 5F YYMM[7:0] YYMM[15:8] DDHH[7:0] DDHH[15:8] MMSS[7:0]
MMSS[15:8] MS[7:0] MS[15:8] SUM

AX~AZ (Acceleration)

Register Name: AX~AZ

Register address: 52~54 (0x34~0x36)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X=AX[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s ²)
15:0	AY[15:0]	Acceleration Y=AY[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s ²)
15:0	AZ[15:0]	Acceleration Z=AZ[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s ²)
Read 3-axis acceleration: 50 03 00 34		

GX~GZ (Angular velocity)

Register Name: GX~GZ

Register address: 55~57 (0x37~0x39)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GX[15:0]	Angular velocity X=GX[15:0]/32768*2000°/s
15:0	GY[15:0]	Angular velocity Y=GY[15:0]/32768*2000°/s
15:0	GZ[15:0]	Angular velocity Z=GZ[15:0]/32768*2000°/s

HX~HZ (Magnetic field)

Register name: HX~HZ		
Register Address: 58~60 (0x3A~0x3C)		
Read and write direction: R		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	HX[15:0]	Magnetic field X=HX[15:0] (unit: LSB)
15:0	HY[15:0]	Magnetic field Y=HY[15:0] (unit: LSB)
15:0	HZ[15:0]	Magnetic field Z=HZ[15:0] (unit: LSB)

Roll~Yaw (Angle)

Register Name: Roll~Yaw		
Register address: 61~63 (0x3D~0x3F)		
Read and write direction: R		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	Roll[15:0]	Roll $X = \text{Roll}[15:0] / 32768 * 180^\circ$
15:0	Pitch[15:0]	Pitch $Y = \text{Pitch}[15:0] / 32768 * 180^\circ$
15:0	Yaw[15:0]	Heading $Z = \text{Yaw}[15:0] / 32768 * 180^\circ$

TEMP (Temperature)

Register Name: TEMP Register Address: 64 (0x40) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature=TEMP[15:0]/100°C

D0Status~D3Status （Port status）

Register Name: D0Status~D3Status Register address: 65~68 (0x41~0x44) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	D0Status[15:0]	D0 state value
15:0	D1Status[15:0]	D1 status value
15:0	D2Status[15:0]	D2 state value
15:0	D3Status[15:0]	D3 state value

PressureL~HeightH （Air pressure altitude）

Register Name: PressureL~HeightH		
Register address: 69~72 (0x45~0x48)		
Read and write direction: R		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	PressureL[15:0]	Air pressure=((int)PressureH[15:0]<<16) PressureL[15:0](Pa)
15:0	PressureH[15:0]	
15:0	HeightL[15:0]	Altitude=((int)HeightH[15:0]<<16) HeightL[15:0](cm)
15:0	HeightH[15:0]	

LonL~LatH (Longitude and latitude)

Register Name: LonL~LatH		
Register address: 73~76 (0x49~0x4C)		
Read and write direction: R		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	LonL[15:0]	Lon[31:0]=((int)LonH[15:0]<<16) LonL[15:0](Pa)
15:0	LonH[15:0]	
15:0	LatL[15:0]	Lat[31:0]=((int)LatH[15:0]<<16) LatL[15:0](cm)
15:0	LatH[15:0]	

The NMEA8013 standard stipulates that the longitude output format of GPS is dd mm. mmmmm (dd is degrees, mm. mmmmm is minutes), and the decimal point is removed from the longitude/latitude output, so the degrees of longitude/latitude can be calculated as follows:

dd=Lon [31:0]/10000000;

dd=Let [31:0]/10000000;

Longitude/latitude fraction calculation:

mm.mmmmm= (Lon [31:0] %10000000)/100000; (% means remainder operation)

mm.mmmmm=(Lat[31:0]%10000000)/100000; (% means remainder operation)

GPSHeight~GPSVH (GPS Data)

Register Name: GPSHeight~GPSVH

Register address: 77~80 (0x4D~0x50)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GPSHeight[15:0]	GPS Altitude=GPSHeight[15:0]/10(m)
15:0	GPSYAW[15:0]	GPS heading=GPSYAW[15:0]/100(°)
15:0	GPSVL[15:0]	GPS ground speed=((int)GPSVH[15:0]<<16) GPSVL[15:0])/1000(km/h)
15:0	GPSVH[15:0]	

q0~q3 (Quaternion)

Register name: q0~q3 Register address: 81~84 (0x51~0x54) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	q0[15:0]	Quaternion0=q0[15:0]/32768
15:0	q1[15:0]	Quaternion1=q1[15:0]/32768
15:0	q2[15:0]	Quaternion2=q2[15:0]/32768

15:0	q3[15:0]	Quaternion3=q3[15:0]/32768
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SVNUM~VDOP (GPS Location Accuracy)

Register Name: SVNUM~VDOP Register address: 85~88 (0x55~0x58) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	SVNUM[15:0]	GPS satellites=SVNUM[15:0]
15:0	PDOP[15:0]	Location longitude=PDOP[15:0]/100
15:0	HDOP[15:0]	Horizontal positioning longitude=HDOP[15:0]/100
15:0	VDOP[15:0]	Longitude of vertical positioning=VDOP[15:0]/100

DELAYT (Alarm signal delay)

Register Name: DELAYT		
Register Address: 89 (0x59)		
Read and write direction: R/W		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	DELAYT[15:0]	Unit: ms After the angle alarm occurs, the port will generate the corresponding alarm signal. When the alarm is removed, the alarm signal will continue to DELAYT[15:0] and then disappear
Example: FF AA 59 E8 03 (set alarm signal delay 1000ms)		

XMIN~XMAX (X-axis angle alarm threshold)

Register Name: XMIN~XMAX		
Register address: 90~91 (0x5A~0x5B)		
Read and write direction: R/W		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	XMIN[15:0]	Set the X-axis angle alarm minimum value $X\text{-axis min} = XMIN[15:0] * 180 / 32768(^{\circ})$
15:0	XMAX[15:0]	Set the X-axis angle alarm maximum value $X\text{-axis max} = XMAX[15:0] * 180 / 32768(^{\circ})$

Example:

FF AA 5A 72 FC (set -5 degrees), 0xFC72=-910,

-910*180/32768=-5 FF AA 5B 8E 03 (set 5 degrees),

0x038E=910, 910*180/32768=5

The X axis will not alarm between -5°~5°, once it exceeds this range, an alarm will occur

BATVAL (Voltage)

Register Name: BATVAL

Register Address: 92 (0x5C)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	BATVAL[15:0]	Voltage=BATVAL[15:0]/100 °C

ALARMPIN (Alarm pin mapping)

Register Name: ALARMPIN

Register Address: 93 (0x5D)

Read and write direction: R/W

Default: 0x4365

Bit	NAME	FUNCTION
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15:12	X-ALARM[15:12]	0001(0x01): D0 0010(0x02): D1 0011(0x03): D2 0100(0x04): D3 0101(0x05): SCL 0110(0x06): SDA
11:8	X+ALARM[11:8]	0001(0x01): D0 0010(0x02): D1 0011(0x03): D2 0100(0x04): D3 0101(0x05): SCL 0110(0x06): SDA
7:4	Y-ALARM[7:4]	0001(0x01): D0 0010(0x02): D1 0011(0x03): D2

		0100(0x04): D3 0101(0x05): SCL 0110(0x06): SDA
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3:0	Y+ALARM[3:0]	0001(0x01): D0 0010(0x02): D1 0011(0x03): D2 0100(0x04): D3 0101(0x05): SCL 0110(0x06): SDA
Example: Set X-alarm signal to output on D3 port Set the X+ alarm signal to output at port D1 Set the Y-alarm signal to output on the SCL port Set the Y+ alarm signal to output at the SCL port Send: FF AA 5D 55 42		

YMIN~YMAX (Y-axis angle alarm threshold)

Register Name: YMIN~YMAX Register address: 94~95 (0x5E~0x5F) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	YMIN[15:0]	Set the Y-axis angle alarm minimum value Y axis angle alarm minimum value=YMIN[15:0]*180/32768(°)
15:0	YMAX[15:0]	Set the Y-axis angle alarm maximum value Y-axis angle alarm maximum value=YMAX[15:0]*180/32768(°)

Example: FF AA 5E 72 FC (set -5 degrees) , 0xFC72=-910,

-910*180/32768=-5 FF AA 5F 8E 03(set 5 degrees) , 0x038E=910,

910*180/32768=5

The Y axis does not alarm between -5°~5°, and an alarm occurs once it exceeds the range

GYROCALITHR (Gyro static threshold)

Register Name: GYROCALITHR

Register Address: 97 (0x61)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15: 0	GYROCALITHR[15:0]	Set the gyroscope static threshold: Gyro static threshold=GYROCALITHR[15:0]/1000(°/s)

Example: Set the gyroscope static threshold to 0.05°/s

FF AA 61 32 00

When the angular velocity change is less than 0.05°/s and lasts for the time of "GYROCALTIME", the sensor recognizes that it is stationary, and automatically resets the angular velocity less than 0.05°/s to zero

The setting rule of the static threshold of the gyroscope can be determined by reading the value of the "WERROR" register. The general setting rule is:
GYROCALITHR=WERROR*1.2, unit: °/s

This register needs to be used in combination with the GYROCALTIME register

ALARMLEVEL (Angle alarm level)

Register Name: ALARMLEVEL Register Address: 98 (0x62) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	ALARMLEVEL[3:0]	To set the alarm level: 0000(0x00): Low level alarm (high level when not alarming, low level when alarming) 0001(0x01): High level alarm (low level when not alarming, high level when alarming)
Example: Setting a high level alarm FF AA 62 01 00		

GYROCALTIME (Gyro auto calibration time)

Register Name: GYROCALTIME Register Address: 99 (0x63) Read and write direction: R/W Default: 0x03E8		
Bit	NAME	FUNCTION
15:0	GYROCALTIME[15:0]	Set gyroscope auto-calibration time

Example: Set gyroscope auto-calibration time to 500ms

FF AA 63 F4 01

When the angular velocity change is less than "GYROCALITHR" and lasts for 500ms, the sensor recognizes that it is stationary and automatically resets the angular velocity less than $0.05^{\circ}/s$ to zero.

This register needs to be used in combination with the GYROCALTIME register

TRIGTIME (Alarm continuous trigger time)

Register Name: TRIGTIME

Register Address: 104 (0x68)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	TRIGTIME[15:0]	Set the alarm continuous trigger time

Example: Set the alarm continuous trigger time to 500ms

FF AA 68 F4 01

When the angle alarm occurs, the alarm signal will not be output immediately, and it will only be output when it needs to last for 500ms

The register is used to filter out alarms caused by malfunctions

KEY (Unlock)

Register Name: KEY

Register Address: 105 (0x69)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	KEY[15:0]	Unlock register: When performing a write operation, you need to set this register first
Example: Unlock, write 0xB588 to this register (other values are invalid) FF AA 69 88 B5		

WERROR (Gyroscope change value)

Register Name: WERROR Register Address: 106 (0x6A) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	WERROR[15:0]	Gyroscope change $\text{value} = \text{WERROR}[15:0] / 1000 * 180 / 3.1415926 (^{\circ}/s)$ When the sensor is stationary, the "GYROCALITHR" register can be set by changing this register.

TIMEZONE (GPS time zone)

Register Name: TIMEZONE Register Address: 107 (0x6B) Read and write direction: R/W Default: 0x0014		
Bit	NAME	FUNCTION

15:8		
7:0	TIMEZONE[7:0]	Set GPS time zone: 00000000(0x0000): UTC-12 00000001(0x0001): UTC-11 00000010(0x0002): UTC-10 00000011(0x0003): UTC-9 00000100(0x0004): UTC-8 00000101(0x0005): UTC-7 00000110(0x0006): UTC-6 00000111(0x0007): UTC-5 00001000(0x0008): UTC-4 00001001(0x0009): UTC-3 00001010(0x000A): UTC-2 00001011(0x000B): UTC-1

		00001100(0x000C): UTC 00001101(0x000D): UTC+1 00001110(0x000E): UTC+2 00001111(0x000F): UTC+3 00010000(0x0010): UTC+4 00010001(0x0011): UTC+5 00010010(0x0012): UTC+6 00010011(0x0013): UTC+7 00010100(0x0014): UTC+8 (Default east 8th district) 00010101(0x0015): UTC+9 00010110(0x0016): UTC+10 00010111(0x0017): UTC+11 00011000(0x0018): UTC+12
Example: FF AA 6B 15 00 (set GPS time zone to East 9)		

WZTIME (Angular velocity continuous stationary time)

Register Name: WZTIME Register Address: 110 (0x6E) Read and write direction: R/W Default: 0x01F4		
Bit	NAME	FUNCTION
15:0	WZTIME[15:0]	Angular velocity continuous rest time

Example: Set the angular velocity continuous static time to 500ms

FF AA 6E F4 01

When the angular velocity is less than "WZSTATIC" for 500ms, the angular velocity output is 0, and the Z-axis heading angle is not integrated

This register needs to be used in combination with the "WZSTATIC" register.